

ASEN 5264 Decision Making under Uncertainty

Exam 2: Online Methods and Reinforcement Learning

Instructions: Clearly indicate your final answers and briefly justify numerical answers with text or mathematical expressions. If you do not understand how to do a problem, skip it and move on so that you have time to attempt all problems. You may consult any static source, but you may NOT communicate with any person except the instructor or TA, and you may not use LLMs such as ChatGPT.

Important: Please include the following statement along with your signature (typed or handwritten) on the exam: **“I affirm that I completed this exam without the use of any artificial intelligence tool such as ChatGPT or Github Copilot.”**

Question 1. (10 pts) Suppose that we are using Q-learning to solve an MDP with $S = \{1, 2\}$ and $A = \{a, b\}$. The current Q values are given in the table below:

s	a	$Q(s, a)$
1	a	5
1	b	3
2	a	4
2	b	2

If we take step $(s = 1, a = a, s' = 2, r = 6)$ and use the Q-learning update rule with $\alpha = 0.1$ and $\gamma = 0.9$, which Q values are changed and what are their new values?

Question 2. (20 pts) Consider a three-armed Bernoulli bandit with win probabilities $\theta = [0.3, 0.7, 0.4]$. After 10 pulls of each arm, the number of wins for each arm is $w = [2, 5, 6]$.

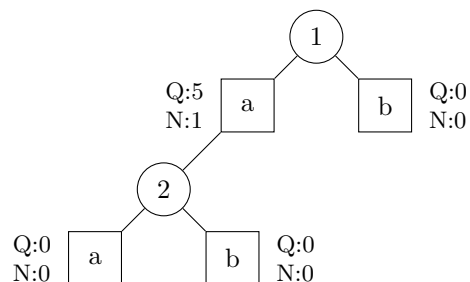
- (a) What is the probability that the next pull will result in a win (+1 reward) if an epsilon greedy policy is used with $\epsilon = 0.1$? Briefly justify your answer.
- (b) What is the probability that the next pull will result in a win (+1 reward) if a UCB policy is used with $c = 1$? Briefly justify your answer.
- (c) After a very large number of pulls, which policy will have the highest average payoff? epsilon-greedy or UCB? Briefly justify your answer.

Question 3. (20 pts) Consider a candidate reinforcement learning algorithm for MDPs with integer states and actions that keeps track of a large table, Q , where each row corresponds to a state and each column to an action. Let $Q[s, a]$ be the entry in row s and column a . The algorithm works as follows:

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1: Initialize  $Q$  to be a matrix of zeros.
2: Initialize  $s$  to be the initial state.
3: loop
4:    $a \leftarrow \operatorname{argmax} Q[s, a]$  (Break ties randomly).
5:   Take action  $a$ , receive reward  $r$ , and observe the next state  $s'$ .
6:    $Q[s, a] \leftarrow Q[s, a] + 0.1(r - Q[s, a])$ 
7:    $s \leftarrow s'$ 
8:   if  $s$  is a terminal state then
9:     Reset  $s$  to the initial state.
10:  end if
11: end loop
```

- (a) Should this algorithm be considered model-based or model-free? Why?
- (b) Should this algorithm be considered on-policy or off-policy? Why?
- (c) What is a major limitation of this algorithm related to exploration? Briefly justify your answer.
- (d) What is a major limitation of this algorithm related to credit assignment? Briefly justify your answer.

Question 4. (20 pts) Consider the following MCTS tree where state nodes are represented with circles and action nodes with squares. States are represented with integers and there are two actions, a and b. If the generative function is called at any point, use the values in the table to the right and use a discount of $\gamma = 0.9$ if needed.



G(s, a)			
s	a	s'	r
1	a	2	6.0
1	b	3	7.0
2	a	4	5.0
2	b	4	6.0

- Suppose we run one iteration of MCTS starting from node 1. What is the path of states and actions that the search phase will take including any new state nodes created (e.g. (1, b, 2, b, 5) or (1, a, 4))? Briefly justify why each node is selected.
- Draw any new state and action nodes that would be created in the expansion phase (you can either add to the existing tree or re-draw the tree).
- Assume the rollout from the new leaf node returns a value estimate of 10. Perform the backup to update the Q and N values at all action nodes nodes on the path (you do not need to copy unchanged values from the existing tree).